

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level)	Marks
Q. 1 Solve <u>any two</u> of the following.		12
A) What is Standard Penetration Test? Explain it in detail?	Understand	06
B) Enlist & explain the geophysical methods of soil exploration?	Remember	06
C) C.1) The following dimensions are given for a Shelby tube sampler: External diameter = 51 mm; Internal diameter = 48 mm. Determine the area ratio?	Apply	06
C.2) 75 mm is the external diameter of sampling tube. If the area ratio required is 20% determine the thickness of sampling tube? In what type of clay would such a high area ratio be required?		
Q.2 Solve <u>any two</u> following questions.		12
A) What are the assumptions in Terzaghi's bearing capacity theory? Explain the bearing capacity equation given by Terzaghi based on general shear failure.	Understand	06
B) A strip footing of width 3 m is founded at a depth of 2 m below the ground surface in a (c - ϕ) soil having a cohesion $c = 30 \text{ kN/m}^2$ and angle of shearing resistance $\phi = 35^\circ$. The water table is at a depth of 5 m below ground level. The moist weight of soil above the water table is 17.25 kN/m^3 . Determine (a) Ultimate bearing capacity of the soil, (b) Net bearing capacity, and (c) Net allowable bearing pressure and the load/m for a factor of safety of 3. Use the general shear failure theory of Terzaghi. $\phi = 35^\circ$ $N_c = 57.8$, $N_q = 41.4$ and $N_\gamma = 42.4$	Apply	06
C) If the water table in Q.2 B) rises to 1.25 m below the ground level and 1.25 m below the base of foundation, determine the net safe bearing pressure of the footing. All the other data given in Q.2 B) remains the same. Assume the saturated unit weight of the soil γ_{sat} is 18.5 kN/m^3 γ (above WT) = 17.5 kN/m^3 .	Evaluate	06

Q. 3	Solve <u>any two</u> of the following.		12
A)	What is expansive soils? Explain in detail the characteristics of expansive soils?	Remember	06
B)	Define and explain the following terms: A) Swelling Potential B) Swelling Pressure C) Free Swell	Remember	06
C)	Explain in detail the IS Code of method for design of raft foundation.	Understand	06
Q.4	Solve <u>any two</u> of the following.		12
A)	Explain in detail the concept of Negative Skin Friction?	Understand	06
B)	Enlist the different methods to estimate the load carrying capacity of piles and explain in detail dynamic formulae method.	Understand	06
C)	C.1) A timber pile was driven by a drop hammer weighing 30 kN with a free fall of 1.2 m. The average penetration of the last few blows was 5 mm. What is the capacity of the pile according to Engineering News Formula? C.2) A pile is driven with a single acting steam hammer of weight 15 kN with a free fall of 900 mm. The final set, the average of the last three blows, is 27.5 mm. Find the safe load using the Engineering News Formula?	Evaluate	06
Q. 5	Solve <u>any two</u> of the following.		12
A)	Enlist the different methods for the analysis of finite slopes and explain in detail the stability number method.	Understand	06
B)	Enlist and explain in detail the different types of slope failures?	Understand	06
C)	An embankment is inclined at an angle of 35° and its height is 15 m. The angle of shearing resistance is 15° and the cohesion intercept is 200 kN/m^2 . The unit weight of soil is 18.0 kN/m^3 . If Taylor's stability number is 0.06, find the factor of safety with respect to cohesion.	Evaluate	06

***** End *****